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Simple and fast near-infrared spectroscopic analysis of hydroxyl number of polyol using a disposable glass vial

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Abstract

Hydroxyl (OH) number of polyol was measured using near-infrared (NIR) spectroscopy with the use of a disposable glass vial as a sample container. Polyols are viscous, so disposable vials are advantageous when spectroscopic methods are employed. Due to the curvature of the vial walls, a narrow aperture was used to minimize the spectroscopic deviations. The narrow aperture attenuated the NIR radiation and increased the spectral noise in the collected polyol spectra. Wavelet transformation (WT) was employed to reduce this noise and partial least squares (PLS) calibration model was developed. The overall prediction results compare well with those from conventional wet analysis that requires time (1–3 h) and large amounts of chemical reagents. NIR spectroscopy with the use of disposable vials can be utilized for a simple and fast quality assurance of polyol in actual industrial settings.
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Keywords: NIR; Near-infrared; PLS; Partial least squares; Wavelet transformation; Disposable vial; Polyol; Polyurethane

1. Introduction

Polyol is an important raw feed material for the manufacture of diverse polyurethane products. It contains multiple hydroxyl (–OH) groups and ranges in molecular weight from 400 to 7000 Da. Usually polyol is produced from reaction of alkylene oxide (ethylene oxide or propylene oxide) and initiator (e.g. ethylene glycol, glycerin) in a necessary catalytic environment. It further reacts with isocyanates (–N=C=O) via the addition polymerization reaction to produce polyurethane products. The determination of hydroxyl (OH) number [1,2] is essential, since it confirms the final product quality. Conventionally, the OH number is measured by a wet chemical analysis [3]. Polyol is refluxed with an excess of acetic anhydride in a non-aqueous medium to esterify hydroxyl groups. Then, excess acetic anhydride is hydrolyzed to acetic acid and titrated with KOH. The same procedure is performed for a blank sample. The difference in acetic acid level between a blank and the sample is the measure of the content of OH groups in the sample. The final OH number is designated by mg KOH that is required to titrate 1 g of polyol (unit: mg KOH/g). This wet analysis requires long analysis time (1–3 h) due to the chemical derivatization and large consumption of chemical reagents such as toxic pyridine.

Near-infrared (NIR) spectroscopy is an excellent analytical method that can replace the conventional wet analysis because it is simple and requires no (or minimal) chemical reagents [4–7]. Polyols are viscous, which makes it difficult to clean the cuvettes and optical fiber probes when NIR spectroscopy is employed. Some researchers have investigated the direct oral care liquid measurement through amber plastic bottle using NIR spectroscopy [8] and determination of residual moisture in lyophilized sucrose through intact glass vials [9]. When a disposable vial is used for polyol analysis, it is necessary to investigate the spectral characteristics of the vial and its influence on the calibration. The vials are often used in industries; however, in-depth spectroscopic studies in relation with the vial have not been followed.

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The OH number of polyol has been determined using NIR spectroscopy with the use of a disposal vial glass, and its spectroscopic characteristics and influence on calibration have been studied. Since glass is fairly transparent in NIR region, no significant radiation loss from the vial itself that can decrease signal-to-noise ratio is expected. However the surface of vial was curved, so a narrow aperture (0.85 mm) was used to minimize the spectroscopic deviations. Therefore, the overall signal-to-noise ratio was degraded. Consequently, the final spectra obtained using the vials were relatively noisier, but without serious distortion of spectral features. Wavelet transformation (WT) [10–12] was utilized to reduce the noise imposed on the raw spectra before performing second derivative algorithm. Finally, partial least squares (PLS) [13,14] calibration model was developed using the resulting second derivative spectra. The NIR prediction results had good correlation with those from wet analysis as well as showed good repeatability even when different vials were used.

2. Experimental

2.1. Sample and data set preparation

Polyol samples were acquired from Dongnam Chemical, Korea. Initially, for the calibration set, 34 samples (hydroxyl number: 21.4–155.1 mg KOH/g) were collected over 4 months to give more process related variations into the data set that would result in the more robust calibration model. After this period, 11 samples were prepared (hydroxyl number: 23.7–154.4 mg KOH/g) over 1 month for the prediction set. All the reference analyses were performed using the wet analysis [3].

2.2. NIR spectra and data processing

All the NIR spectra were collected using an ABB Bomem MB-160 FT-NIR spectrometer (Quebec City, Quebec, Canada) equipped with a tungsten–halogen source and DTGS detector. The vial holder was used to hold a glass vial as well as control the temperature. The narrow rectangular aperture of 0.85 mm (width) was used to minimize the spectroscopic deviations from the round surface of a vial. NIR spectra were collected over 9500–4500 cm^{-1} range with the resolution of 8 cm^{-1}. Two spectra were collected for each sample and the temperature of sample was controlled at 75 °C. Air was used as a background for the whole samples.

All the algorithms including PLS regression and preprocessing methods were accomplished using GRAMS/32 software with add-on PLS algorithm (Galactic Industries, Salem, NH, USA). The Wavelet transformation was accomplished using Matlab Version 6.5 (The MathWorks, MA, USA).

3. Results and discussion

3.1. Overall spectral features of polyol

Fig. 1 shows the entire NIR spectral features (9500–4500 cm^{-1}) of three selected polyol samples (OH number: 89.5, 122.8 and 155.1 mg KOH/g) using disposable vials. It is well known that there are generally four major hydrocarbon absorption bands (two combinations, first and second overtone) over the NIR region. Both the molar absorptivity and spectral selectivity decrease as the absorption bands move to the shorter wavelength region (further from the fundamental vibration in the MIR region). The 4700–4000 (not shown in Fig. 1), 6300–5200, 7500–6400 and 9100–7800 cm^{-1} bands correspond to the combination, first overtone, combination and second overtone bands of the CH bond, respectively. Bands centered at 6900 and 4800 cm^{-1} are OH combination and overtone absorptions, respectively. As shown, the absorptions of both hydroxyl bands increase with the OH number.

The spectral sensitivity of the 6000–4500 cm^{-1} region is superior; however, it should not be used for calibration due to the strong sample absorption (up to 2.1 absorbance from the 6.11 mm optical pathlength). The 7500–6000 cm^{-1} range was optimal to use for calibration since the spectral variations based on the change of hydroxyl content are clearly present, while the overall absorbance was under 1.0 absorbance unit to avoid the possible nonlinearity in a calibration model due to deviations from Beer's Law. Consequently, for the calibration, the 9100–6000 cm^{-1} range was used to include the 9100–7500 cm^{-1} (hydrocarbon bands) as well as 7500–6000 cm^{-1} range (hydroxyl information).

3.2. Spectral features of vials

There are two issues to consider when a disposable vial is used as a sample container. The first is the cylindrical shape of the vial that could distort the NIR spectra, and the second is the variability of the disposable vials that influences

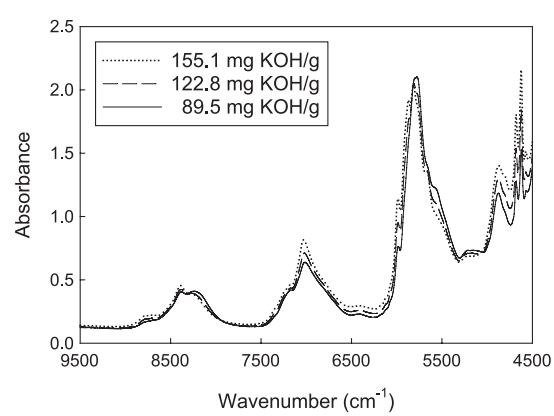

Fig. 1. NIR spectral features (9500–4500 cm^{-1}) of selected three polyol samples (OH number: 89.5, 122.8 and 155.1 mg KOH/g) using disposable vials.

reproducibility. The use of vial should be critically considered for each individual application. If the spectral variations from these two parameters are negligible compared to the spectral variations from the hydroxyl content, the disposable vial can be employed.

To examine whether there were significant spectral differences between normal (flat) and round surface cells, NIR spectra of empty glass vials and rectangular glass cells were compared. The pathlength for the rectangular cell was controlled to have the same pathlength as the vial (6.11 mm). The resulting spectra are shown in Fig. 2. The vial spectrum was acquired with the narrow aperture to minimize the spectroscopic deviations, as described in Section 2. In both spectra, the characteristic glass band at 7100 cm^{-1} was identified and no significant spectral differences were observed. The vial spectrum was relatively noisier because of the attenuation of NIR radiation from the narrow aperture. This result supports that the glass vial can be adopted as a sample container without serious distortion of spectra, but with the lower signal-to-noise ratio.

Using 20 different vials (randomly selected), the reproducibility of vial conditions was examined. Outer diameter (O.D.), inner diameter (I.D.) and thickness of each vial were measured. Then, mean, standard deviation and relative standard deviation (RSD) were calculated as shown in Table 1. Based on the table, outer diameters of vials are well controlled with RSD of 0.37%. The variation of inner diameters (the pathlength of measurement) is slightly larger but still reproducible with RSD of 1.37%. A comparison of the spectra of 20 different empty vials revealed no significant differences, although the baselines varied slightly due to refraction and scattering.

3.3. PLS models

PLS [13,14] regression was used to build a calibration model. In the application of the PLS algorithm, it is generally known that the spectral range and the number of PLS factors are important parameters [15]. The spectral range determines the location of necessary spectral information and the number of PLS factors should be optimally selected to avoid an overfitting. In this research, only the 9100–6000 cm^{-1} range was used to include the hydrocarbon and hydroxyl information.

Table 1
Mean, standard deviation and RSD of outer diameter, inner diameter and thickness from 20 different vials

	Outer diameter (mm)	Inner diameter (mm)	Thickness (mm)
Mean	8.14	6.11	2.03
Standard deviation	0.03	0.08	0.08
RSD (%)	0.37	1.37	3.99

The low signal-to-noise ratio due to narrow aperture and the baseline variation must be considered to optimize the calibration. To reduce baseline variation, the second derivative algorithm was used; however, this tends to over-emphasize the spectral noise. Fig. 3 (lower spectrum) shows the second derivative spectrum of one polyol sample (OH number: 122.8 mg KOH/g). It clearly shows complex and misleading peaks due to noise, even though the spectral variations of the raw spectrum are simple. Therefore, it is necessary to reduce the noise superimposed on the polyol spectra before performing the second derivative algorithm.

WT [10] was used to selectively reduce the noise from the raw spectra. This method decomposes an original spectrum using a wavelet, that is the projection onto a wavelet basis function. Wavelet coefficients of the decomposed spectrum can be manipulated for the various purposes such as de-noising or signal compression. Generally, the de-noising procedure involves removing the specific wavelet coefficients associated with noise and then inversely transforming back to the original domain. In this study, discrete wavelet transformation (DWT) using Daubechies wavelet (order: 5, level: 5) was employed. The details of DWT are described in several articles [11,12]. As the result of DWT, two sets of wavelet coefficients are obtained: detail (d^n) and approximation coefficients (a^n). The superscript n is the level of resolution. The detail and approximation coefficients can be viewed as the result of high-pass and low-pass

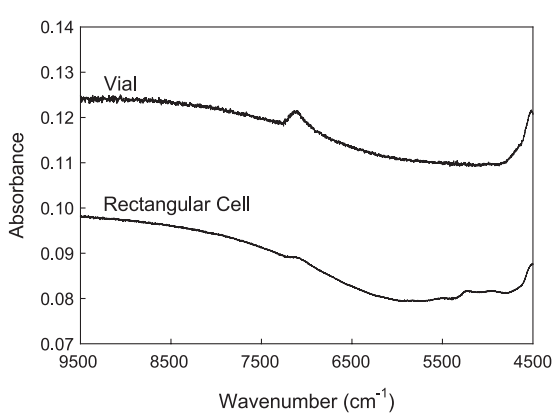

Fig. 2. NIR spectra of empty vial and rectangular glass cell.

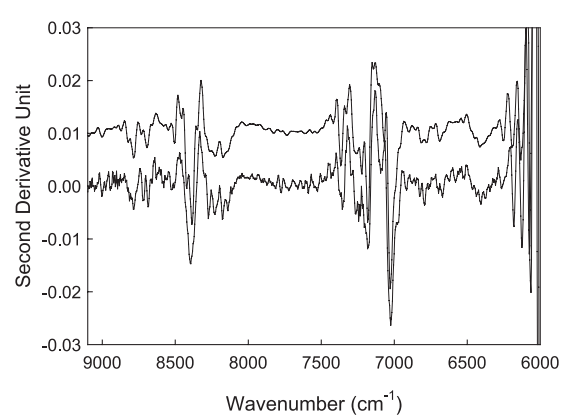

Fig. 3. Second derivative spectra using the raw spectrum (lower) and de-noised spectrum by DWT (upper).

filtering. Usually, the detail coefficients are regarded as noise, especially in NIR spectroscopy where absorption bands are naturally broad. Fig. 4(a) shows the three-level detail coefficients (d^1, d^2, d^3) that are presumed to be spectral noise. These selected detail coefficients were removed before the back transformation to the original spectral domain. The removal of further detail coefficients (such as d^4, d^5) resulted in the distortion of spectral shape. The resulting de-noised spectrum is shown also in Fig. 4(b) and compared to the spectrum without DWT. The de-noised spectrum is offset for better visual comparison. There is no significant difference between original and de-noised spectra in general; however, as shown in the inside of Fig. 4(b), it is clear that the spectral noise is decreased.

Fig. 3 shows the second derivative spectrum (upper spectrum) using the de-noised spectrum. This spectrum is offset for the better visual comparison. The resulting second derivative spectrum is simpler with reduced noise information while retaining the necessary analytical information. Since WT effectively reduces high frequency noise, the second derivative spectra after WT were used to build the PLS calibration models.

The optimum number of factors was identified as the one that gave a minimum standard error of cross validation (SECV). The cross validation was used as a validation method by dividing the calibration data set into four segments. The SECV is an useful index to find the optimal number of PLS factors without overmodelling. The SECV was plotted as a function of the number of PLS factors and its

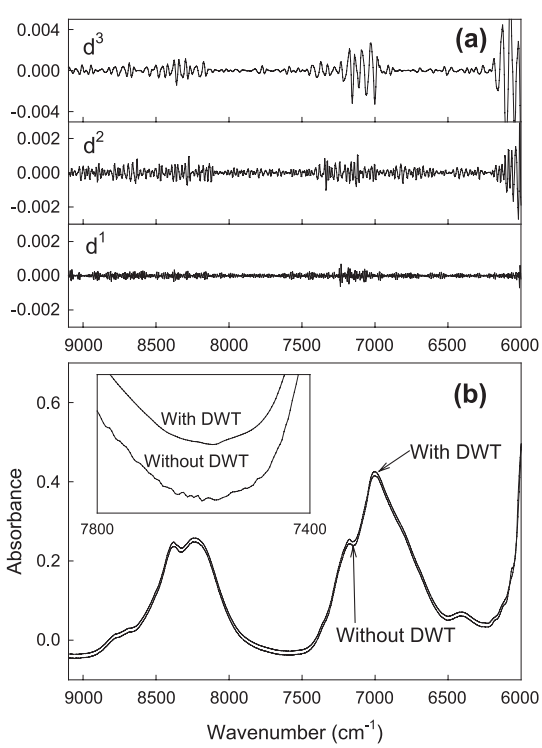

Fig. 4. Three level detail wavelet coefficients (a) and spectra with and without DWT (b).

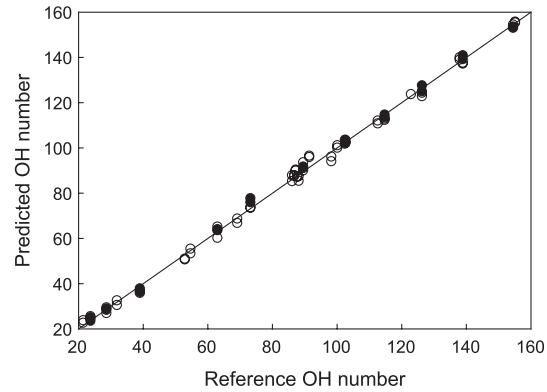

Fig. 5. Scatter plot showing correlation between reference and NIR analyses. Open and filled circles represent the calibration and prediction data, respectively.

decrease pattern was examined. The SECV decreased sharply for the first factor, gradually for the second factor, but after the number of factors of 5, no significant improvement in SECV was observed. With five factors the data have been over-fit by incorporating spectral information into the model that is not related to the hydroxyl content. In this case, four PLS factors were chosen as the optimum. The achieved SECV was 2.14 mg KOH/g. The standard error of prediction (SEP) was calculated by predicting the OH number of samples in the prediction set that were independently collected as described in Section 2. The SEP and mean percent error (MPE) were 2.26 mg KOH/g and 1.7%, respectively. The corresponding scatter plot showing correlation between reference and NIR analyses is shown in Fig. 5. Open and filled circles represent the calibration and prediction data, respectively. The calibration and prediction data correlate well with the reference data.

3.4. Repeatability of prediction using different vials

To test the variation of actual prediction when different vials were used, the same sample was measured 20 times using different vials. The collected 20 spectra were predicted using the calibration model described in the previous section. This procedure was performed on two different polyol samples (85.9 and 96.4 mg KOH/g). The results are summarized in Table 2. The standard deviations of two samples are similar to each other (around 0.4 mg KOH/g) and relatively small compared to the measurement range. When different vials are used at each measurement, the prediction variation will not be significant.

Table 2
The average and standard deviation of 20 predicted values for two samples (85.9 and 96.4 mg KOH/g)

OH number	Average	Standard deviation
Unit: mg KOH/g		
85.9	86.0	0.42
96.4	97.1	0.40

4. Conclusion

The hydroxyl number of polyol was simply measured within 2 min using NIR spectroscopy combined with an inexpensive disposable glass vial. The narrow rectangular aperture, used to minimize the spectroscopic deviations, significantly attenuated the intensity of NIR radiation and led to lower signal-to-noise ratio. To achieve sufficient accuracy on the lower hydroxyl number range, signal-to-noise ratio of the spectra must be improved even though the advanced de-noising algorithms are simultaneously utilized. If the aperture is increased, the signal-to-noise ratio will be improved; however, the reproducibility of spectra is degraded. Future work will involve improving the signal-to-noise ratio by various approaches such as using the high-power light source as well as more sensitive detector.

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References

[1] S. Sigga, Quantitative Organic Analysis Via Functional Groups, John Wiely and Sons, New York, 1949.
[2] R.B. Roy, C. Kradjel, J. Polym. Sci., A, Polym. Chem. 26 (1988) 1733.
[3] American Society for Testing and Materials, Annual Book of ASTM Standards (Method D 2848), Philadelphia, 1982.
[4] D.A. Burns, E.W. Ciurczak, Handbook of Near-Infrared Analysis, Marcel Dekker, New York, 1992.
[5] K. Yamamoto, R. Kakita, C. Suzuki, M. Meno, Y. Nozawa, Appl. Spectrosc. 57 (2003) 353.
[6] K.A. Bunding Lee, R.W. Chylla, T.E. Janota, Appl. Spectrosc. 47 (1993) 94.
[7] W.G. Hansen, Appl. Spectrosc. 47 (1993) 1623.
[8] N.W. Broad, R.D. Jee, A.C. Moffat, M.J. Eaves, W.C. Mann, W. Dziki, Analyst 125 (2000) 2054.
[9] M.S. Kamat, R.A. Lodder, P.P. Deluca, Pharm. Res. 6 (1989) 961.
[10] C.K. Chui, Introduction to Wavelets, Academic Press, Boston, 1991.
[11] B. Walczak, D.L. Massart, Chemom. Intell. Lab. Syst. 36 (1997) 81.
[12] X. Shao, A.K. Leung, F. Chau, Acc. Chem. Res. 36 (2003) 276.
[13] H. Martens, T.M. Naes, Multivariate Calibration, John Wiley and Sons, New York, 1989.
[14] K.R. Beebe, R.J. Pell, M.B. Seasholtz, Chemometrics: A Practical Guide, John Wiley and Sons, New York, 1998.
[15] H. Chung, J.S. Lee, M.S. Ku, Appl. Spectrosc. 52 (1998) 885.